

ComicCraft - AI Comic Story Creator using Gemini Models

8. Project Summary and Future Scope

Technology Track: Generative AI with Google

1. Project Summary

ComicCraft is an AI-powered web application, built for the Generative AI with Google track, that turns a plain-text story idea into a complete, illustrated, multi-panel comic strip. It combines Google Gemini's text generation for the story script and dialogue with Gemini's image generation for matching artwork, producing a consistent visual result from nothing more than a short written idea.

2. What Was Achieved

- Built a working end-to-end pipeline from story prompt to finished comic strip.
- Integrated both Gemini text and image generation models into a single application.
- Maintained visual consistency for characters across separately generated panel images.
- Delivered typical results in under a minute, with graceful handling of errors and invalid input.
- Supported panel-level regeneration and export of the finished comic.

3. Technology Stack Used

Layer	Technology
Frontend	React.js, HTML5, CSS3, JavaScript
Backend	Node.js, Express.js
AI Models	Google Gemini API — text and image generation
Storage (optional)	Firebase Firestore
Hosting	Firebase Hosting / Google Cloud Run
Version Control	Git and GitHub

4. Key Learnings

- How to chain a text generation model and an image generation model into one coherent creative pipeline.
- The importance of a fixed style/character description for keeping AI-generated art visually consistent.
- Practical techniques for handling AI API errors and rate limits gracefully in a user-facing product.

5. Limitations

- Only a small starter set of art styles is currently available.
- Very long story prompts may need trimming for the cleanest panel breakdown.
- Final art quality is bounded by the capability of the Gemini image model used during development.

6. Future Scope

- Add more art styles, such as manga, watercolor, or noir, for greater creative variety.
- Allow users to upload a reference character image for even tighter visual consistency.
- Add multi-language support for captions and dialogue.
- Introduce a community gallery where users can browse and share generated comics.
- Explore turning finished comics into short animated sequences as a future extension.

7. Conclusion

ComicCraft successfully demonstrates how Google Gemini's text and image generation can be combined into a single, practical pipeline, turning a short written idea into a finished, illustrated comic strip — making creative visual storytelling accessible to anyone, with clear directions for future improvement.